ECE 308 Lab 2: Modeling the Rotary Servo Motor

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1. Introduction

This lab's goal is to develop a model of the servo motor to have a basis for controller designs for the following labs. The primary finding from this experiment is the derivation of a relationship between the speed of the motor and the input voltage. This lab goes through three ways to demonstrate system modeling. The first is first principles, which are done by derivations based on physical properties and electrical and mechanical system models. The second method is step responses. This is done by first deriving the expression for the step response of the motor, finding the steady state value, then calculating the parameters. The third method is frequency response, which is done by recording the relationship between frequency changes and the steady state gain.

2. Results

Step Response Tests

i. Provide the expression for the step response for the motor, in terms of the size of the input step (Av) and the motor parameters a0 and b0.

(BoAv/ao)(1-e^(-a0t))

ii. Provide the measurements that you took in the lab (i.e., Tables 2.1 and 2.2).

Steady State Value	3.247
Time Constant	28.404ms

Table 2.1: Measured Parameters from step response

1/time constant = a0

1/0.0284 = 35.206

A = b0 * Av / a0

B0 = A * a0 / Av = 3.247 * 35.206 / 2 = 58.025

a0	35.206	b0	58.025

Table 2.2: Calculated parameters for the motors based on the step response

iii. Provide the plot of the step response that you obtained. On your plot, indicate the steady state value and the point where the step response reaches (1 − e−1) times its steady state value.

Figure 2.1: Step Response Measuring Steady State Value

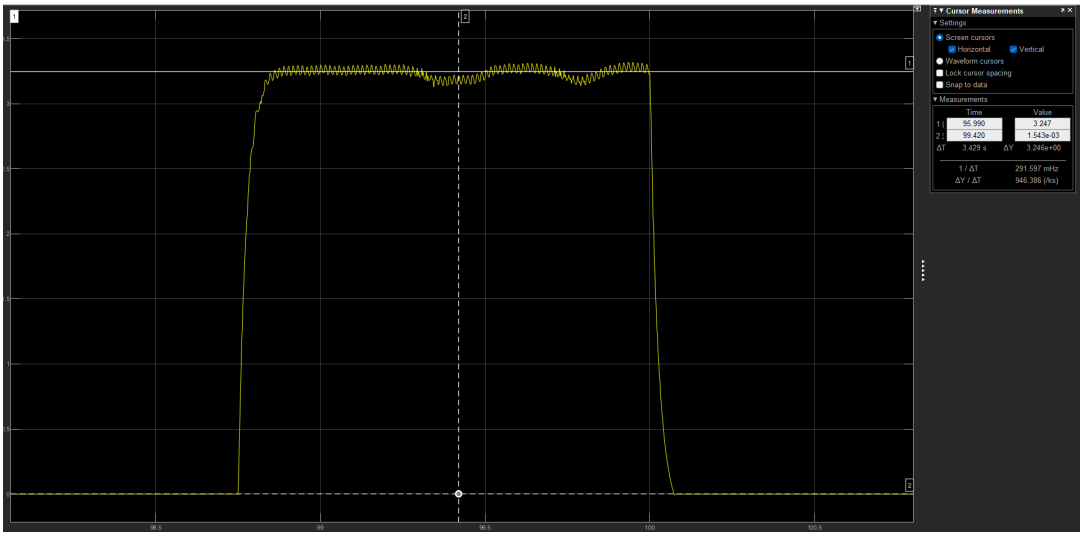
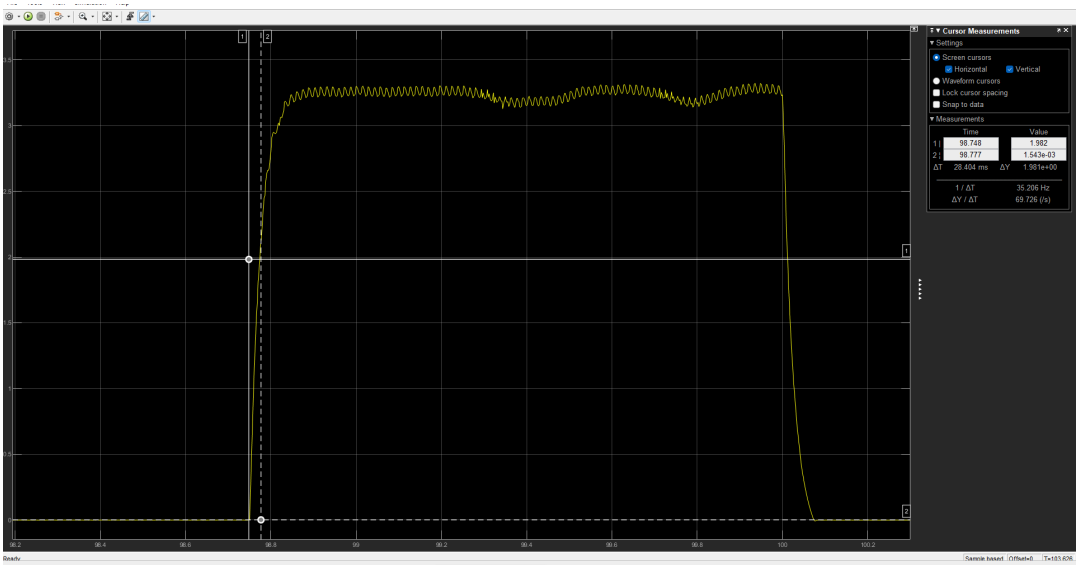


Figure 2.2: Step Response Measuring Time Constant



The Steady State response of the function is 3.247V. The step response reaches (1- e^-1) its steady state value 28.404ms after it turns on.

Frequency Response Tests:

i. Provide the expression for the amplitude of the output sinusoid for the motor, in terms of the amplitude of the input sinusoid (A), the input frequency ω, and the motor parameters a0 and b0.

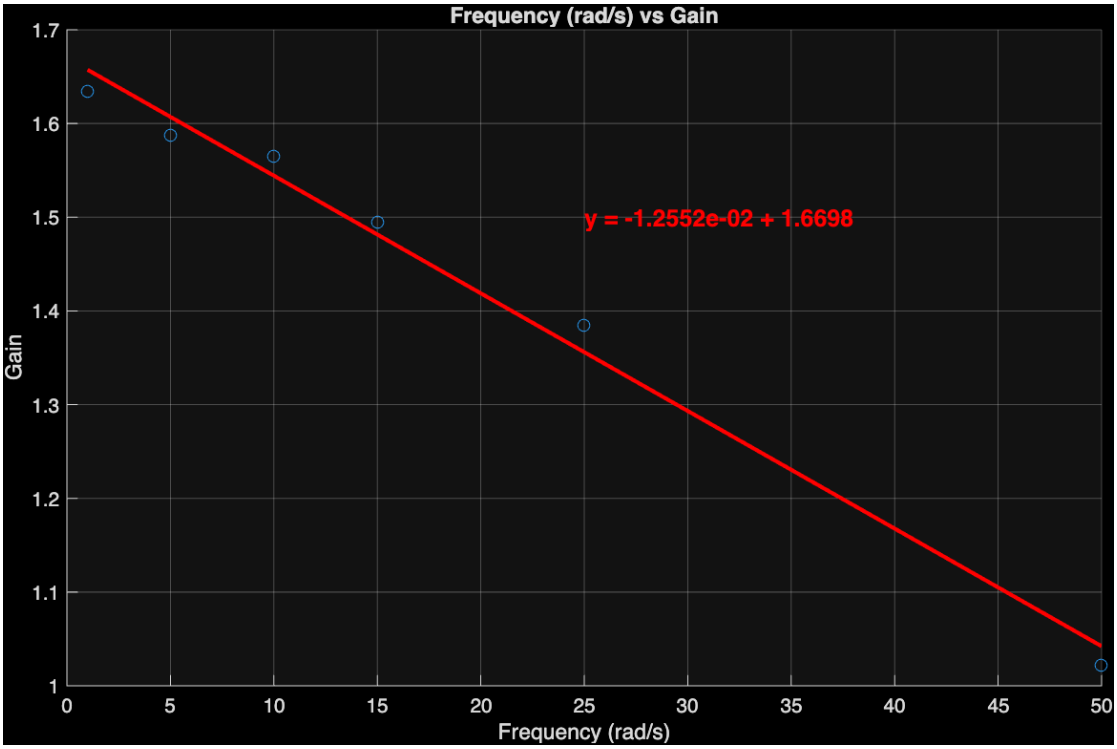
A = bo/sqrt(w^2+ao^2)

ii. Provide the measurements that you took in the lab (i.e., Tables 2.3 and 2.4).

Frequency (rad/s)	Amplitude of Input	Amplitude of Output	Gain
1	2	3.268	1.634
5	2	3.170	1.5875
10	2	3.130	1.565
15	2	2.990	1.495
25	2	2.770	1.385
50	2	2.044	1.022

Table 2.3: Measured Parameters and Frequency Response

iii. Provide the plot of the best-fit line that you drew in order to estimate the parameters a0 and b 0 from the data.



A = b0/sqrt(w^2+a0^2)

A^2 = b0^2 / (w^2 + a0^2)

1/A^2 = w^2/b0^2 + a0^2/b0^2

A = m * sqrt(w) + b

Gain = 1.67 – 0.012552(w)

Gain = 1/sqrt(2) * 1.67 = 1.18

1.18 = 1.67 – 0.012552(w)

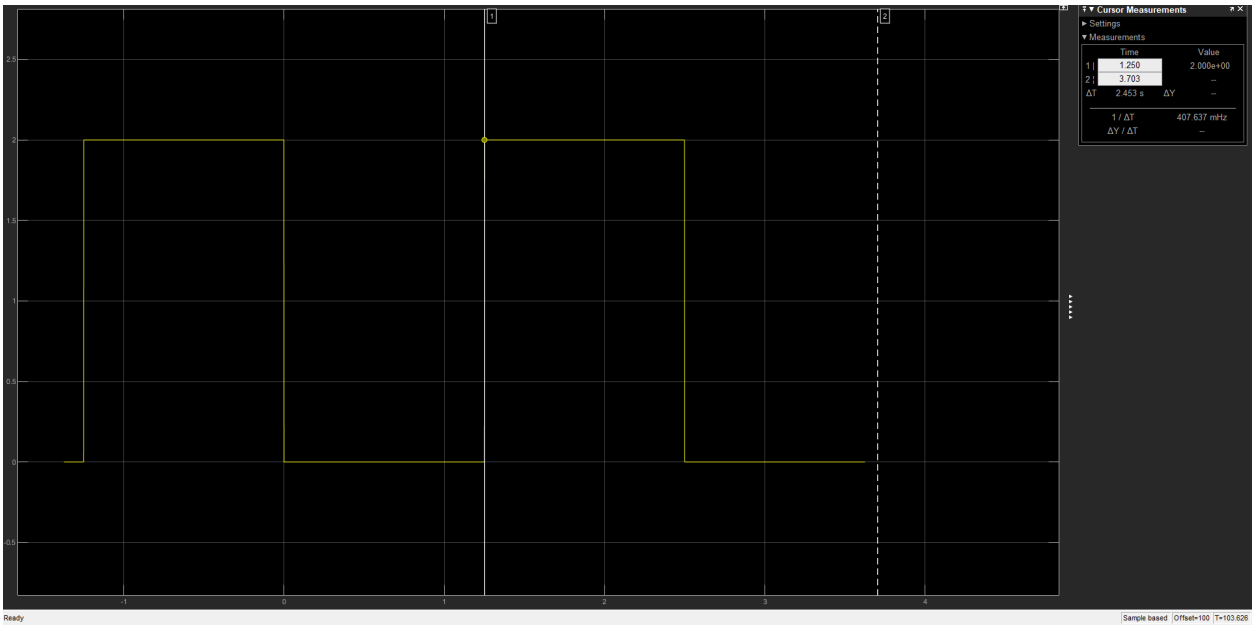
W = 39.02 = a0

B0 = 39.02 * 1.67 = 65.15

a0	39.02	b0	65.15

Table 2.4: Calculated parameters for the motors based on the frequency response

➔ (-1.2552e-02)w + 1.6698 = b0/sqrt(w^2 + a0^2)



3. Conclusions

Both tests appeared to output similar numbers, with roughly only 10-15% difference between the found numbers. This is indicative that both tests were accurate and are valid ways of finding a0 and b0. Both step response tests and frequency response tests are reliable ways of finding the a0 and b0 coefficients, which makes sense, as this system is defined by a single differential equation that defines the relationship between input voltage and output speed. While the tests themselves are different, with the step response measuring the transient behavior, and the frequency response measuring the steady state behavior, they are both measuring the motor's internal physics, which remain the same.